

Practice Questions

Module: Java Programming Fundamentals

Section A: Command Line Arguments

Problem 1: Company Name Generator

A software company wants to automatically generate its official office name. Develop a Java program that accepts the **company name** and **city name** as command-line arguments and displays the office name in the following format:

Output Format

<Company Name> Technologies <City Name>

Sample Input

ABC Delhi

Sample Output

ABC Technologies Delhi

Problem 2: Personalized Welcome Message

Develop a Java application that accepts the user's name as a command-line argument and displays a personalized welcome message.

Sample Output

Welcome Rahul

Problem 3: Simple Addition Utility

Develop a Java program that accepts two integer values through command-line arguments and displays their sum in a meaningful sentence.

Sample Output

The sum of 25 and 45 is 70.

Section B: Decision Making (if-else)

Problem 4: Number Category Identifier

Design a Java program that determines whether a given integer is:

- Positive
- Negative
- Zero

Display the appropriate message.

Problem 5: Even or Odd Checker

Develop a program that determines whether an entered integer is **Even** or **Odd**.

Problem 6: Command-Line Validator

Develop a Java program that verifies whether command-line arguments are provided.

- If no arguments are supplied, display:

No Values

- Otherwise, print all the arguments separated by commas.
-

Problem 7: Character Sorter

Develop a Java program that accepts two characters and prints them in alphabetical order.

Example

Input:

z a

Output:

a z

Problem 8: Character Classifier

Design a Java application that identifies whether a given character is:

- Alphabet
- Digit
- Special Character

Display the corresponding category.

Problem 9: Bank Interest Calculator

A bank offers different interest rates based on the customer's gender and age.

Develop a Java program that accepts:

- Gender
- Age

and displays the applicable interest rate according to the given criteria.

Problem 10: Alphabet Case Converter

Develop a Java application that accepts an alphabet and converts:

- Uppercase to Lowercase
- Lowercase to Uppercase

Display both the original and converted characters.

Section C: Switch Statement

Problem 11: Traffic Signal Color Identifier

Develop a Java program that accepts a color code and displays its corresponding color.

Code	Color
------	-------

R	Red
---	-----

Code Color

G Green

B Blue

Y Yellow

O Orange

W White

Display "**Invalid Code**" for any other input.

Problem 12: Month Name Finder

Develop a Java application that accepts a month number (1–12) and displays the corresponding month name.

If the entered month number is invalid, display an appropriate error message.

Section D: For Loop

Problem 13: Number Sequence Generator

Develop a Java program that prints numbers from **1 to 10** in a single row separated by tabs.

Problem 14: Even Number Printer

Develop a Java application to display all even numbers between **23 and 57**.

Each number should appear on a separate line.

Problem 15: Prime Number Checker

Develop a Java program that determines whether a given integer is a prime number.

Problem 16: Prime Number Generator

Generate and display all prime numbers between **10 and 99**.

Problem 17: Digit Sum Calculator

Develop a Java program that calculates the sum of all digits in a given integer.

Example

Input

4567

Output

22

Problem 18: Floyd's Triangle

Develop a Java program that prints Floyd's triangle using the '*' symbol for the number of rows specified by the user.

Example (4 rows)

```
*  
* *  
* * *  
* * * *
```

Section E: While Loop

Problem 19: Number Reverser

Develop a Java application that reverses the digits of a given integer.

Example

Input

12345

Output

Problem 20: Palindrome Number Checker

Develop a Java program that checks whether a given integer is a palindrome.

Display an appropriate message.

Problem 21: Common Multiple Generator

Develop a Java program that prints the first five numbers divisible by **2, 3, and 5** simultaneously.

Problem 22: Menu-Driven Calculator

Design a menu-driven calculator using a **while loop**.

The program should repeatedly display the following options:

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

After performing the selected operation, ask the user whether to continue.

Continue execution if the user enters **Y** or **y**; otherwise, terminate the program.

Module: Arrays in Java

Section A: One-Dimensional Arrays

Problem 1: Array Statistics Calculator

Develop a Java program to initialize an integer array with user-defined values. The program should calculate and display:

- The sum of all elements
- The average of the elements

Learning Outcome: Understand array initialization, array traversal, and basic arithmetic operations using one-dimensional arrays.

Problem 2: Maximum and Minimum Finder

Develop a Java program to initialize an integer array and determine the largest and smallest elements present in the array.

Learning Outcome: Apply array traversal techniques to identify extreme values within a dataset.

Problem 3: Linear Search Implementation

Design a Java application that initializes an integer array and searches for a specified element using the Linear Search technique.

If the element is found, display its index position; otherwise, display **-1**.

Sample Input

Array: {1, 4, 34, 56, 7}
Search Element: 90

Sample Output

-1

Sample Input

Array: {1, 4, 34, 56, 7}
Search Element: 56

Sample Output

3

Learning Outcome: Implement linear search using one-dimensional arrays.

Problem 4: ASCII to Character Converter

Develop a Java program that initializes an integer array with ASCII values and displays the corresponding characters in a single line.

Sample Input

65 66 67 68 69

Sample Output

A B C D E

Learning Outcome: Understand the relationship between integer values and character encoding.

Problem 5: Largest and Smallest Elements

Develop a Java application to identify:

- The largest two elements
- The smallest two elements

from a given integer array.

Learning Outcome: Practice comparison techniques and array traversal.

Problem 6: Array Sorting

Design a Java program to arrange the elements of an integer array in ascending order and display the sorted array.

Learning Outcome: Implement sorting logic using arrays.

Problem 7: Duplicate Element Removal

Develop a Java program that removes duplicate values from an integer array and displays the resulting array containing only unique elements.

Sample Input

12 34 12 45 67 89

Sample Output

12 34 45 67 89

Learning Outcome: Learn techniques for identifying and eliminating duplicate values from arrays.

Problem 8: Most Frequent Element

Develop a Java application to determine the element that occurs most frequently in an integer array.

Sample Input

10 20 10 30 40 100 99

Sample Output

10

Learning Outcome: Apply frequency counting techniques using arrays.

Problem 9: Conditional Array Sum

Develop a Java program to calculate the sum of all array elements with the following condition:

- If the array contains a **6**, ignore the value **6**, the value **7**, and all elements occurring between them during the sum calculation.
- If no such sequence exists, calculate the normal sum.

Sample Input 1

10 3 6 1 2 7 9

Sample Output

22

Explanation

$10 + 3 + 9 = 22$

Sample Input 2

7 1 2 3 6

Sample Output

19

Sample Input 3

1 6 4 7 9

Sample Output

10

Learning Outcome: Apply conditional logic while traversing arrays.

Section B: Two-Dimensional Arrays

Problem 10: Reverse a 2×2 Matrix

Develop a Java program that accepts **four integer values** as command-line arguments and stores them in a **2×2 matrix**.

The program should:

1. Validate that exactly four integers have been provided.
2. Display the original matrix.
3. Display the matrix in reverse order.

Sample Input

java Sample 1 2 3

Sample Output

Please enter exactly 4 integer numbers.

Sample Input

java Sample 1 2 3 4

Sample Output

The given matrix is:

1 2
3 4

The reversed matrix is:

4 3
2 1

Learning Outcome: Understand matrix representation and traversal in reverse order.

Problem 11: Largest Element in a 3×3 Matrix

Develop a Java application that accepts **nine integer values** as command-line arguments and stores them in a **3×3 matrix**.

The program should:

1. Validate the number of command-line arguments.
2. Display the matrix.
3. Identify and display the largest element present in the matrix.

Sample Input

```
java Sample 1 2 3
```

Sample Output

```
Please enter exactly 9 integer numbers.
```

Sample Input

```
java Sample 1 23 45 55 121 222 56 77 89
```

Sample Output

```
The given matrix is:
```

```
1    23    45
55 121 222
56  77   89
```

```
The largest element in the matrix is: 222
```

Learning Outcome: Develop the ability to process two-dimensional arrays and perform matrix-based computations.

Java basic program for practice

1. **Hello World Program**
Write a Java program to print "Hello, World!" to the console.
2. **Sum of Two Numbers**
Write a Java program that takes two integers as input and outputs their sum.
3. **Factorial of a Number**
Write a Java program to calculate the factorial of a given number.
4. **Fibonacci Series**
Write a Java program to print the Fibonacci series up to a given number of terms.
5. **Palindrome Check**
Write a Java program to check if a given string is a palindrome.
6. **Prime Number Check**
Write a Java program to check if a given number is a prime number.
7. **Reverse a Number**
Write a Java program to reverse the digits of a given number.
8. **Leap Year Check**
Write a Java program to check if a given year is a leap year.
9. **Armstrong Number Check**
Write a Java program to check if a given number is an Armstrong number.
10. **Simple Calculator**
Write a Java program to perform basic arithmetic operations (addition, subtraction, multiplication, division) based on user input.